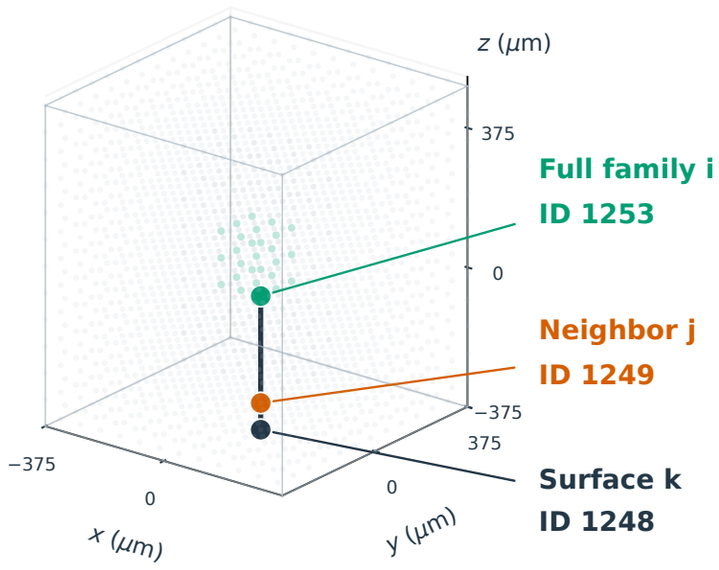
Neighbor dependence and inward force distribution

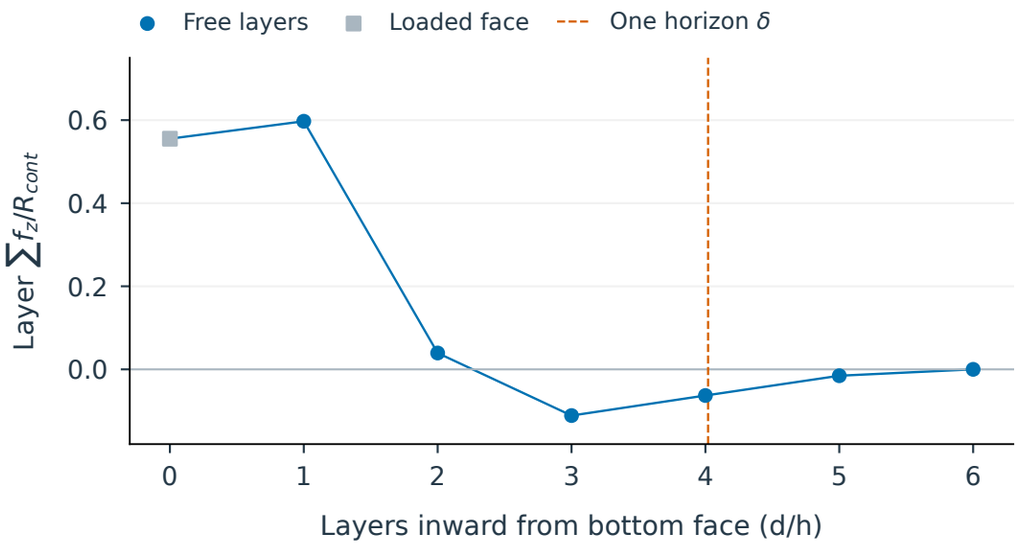
Actual particle links, saved layer forces, and exact force decomposition

(a) Actual particles in three dimensions

Geometry displacement ×50; coordinates in μm

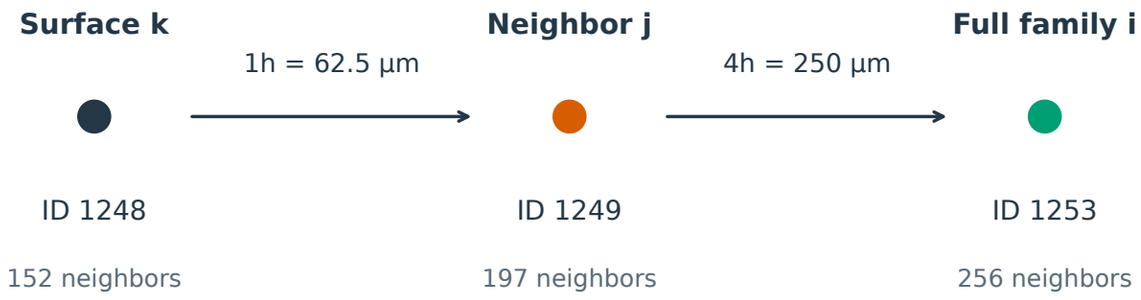


(b) Signed lower-half layer forces



Layer 5: signed sum / R = -0.01534; depth = 5h > δ

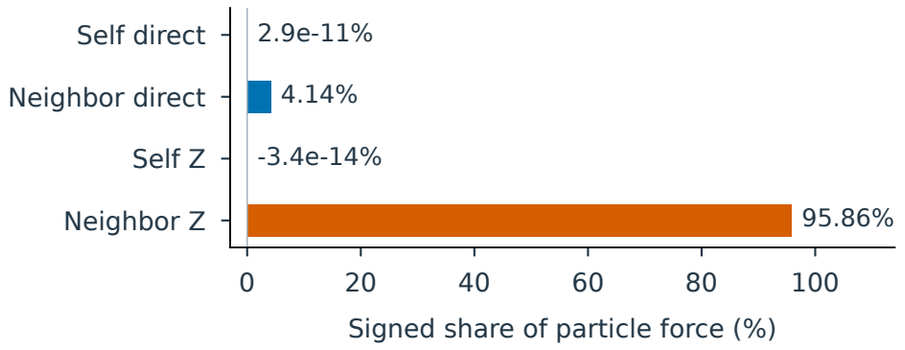
(c) One actual dependency path



Reference link lengths; diagram spacing is schematic.

(d) Force decomposition at particle 1253

Saved axial force: -0.361982 nN



Neighbor Z sums over all neighbors, not just this path.

L075 | 2,197 particles | Strain 0.3% | h = 62.5 μm | δ = 251.25 μm

Arrows indicate coefficient dependence; these static data do not establish a time-evolving propagation front.